

Euler's identity and exponential form

Key Points

Euler's identity states

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Hence a complex number may be written in exponential form

$$z = |z|e^{i \operatorname{Arg} z}$$

Remember that the argument should be between $-\pi$ and π

$$-\pi < \operatorname{Arg} z \leq \pi$$

Basic Skills

Q1. a) Find the modulus and argument of the following complex numbers

i) $z = 2 + 3i$

ii) $w = 3i - 2$

iii) $u = -z$

iv) $v = -w$

b) Convert the complex numbers z, w, u, v into modulus, argument form

c) Convert the complex numbers z, w, u, v into exponential form

d) Plot the complex numbers z, w, u, v

Competency Skills

Q2. Show that the famous formula $e^{i\pi} + 1 = 0$ holds

Q3. Convert the following exponential complex numbers into cartesian form, $a + bi$

a) $2e^{\frac{\pi i}{3}}$ b) πe^{2i} c) $\frac{e^{\frac{\pi i}{5}}}{3}$

Q4. State the modulus and argument of the following complex numbers and plot them on an argand diagram

a) $e^{\frac{2\pi i}{3}}$ b) $5e^i$ c) $\frac{5e^{-\frac{3\pi i}{4}}}{9}$

Q5. For the complex numbers $z = 3e^{\frac{\pi i}{3}}$ and $w = 4e^{-\frac{\pi i}{2}}$ compute the following

a) zw b) $\frac{z}{w}$ c) $z + w$ d) $z - w$

Advanced Skills

Q7. Someone suggests that they have plotted the complex number $z = -5e^{\frac{5\pi}{2}}$, decide if it is possible to have such a complex and if so plot it.

Q8. For the complex number $z = 3 - 4i$ compute the following

a) z^3

b) $z^{10} + 2i - 5$

c) $z^5(2 + i)^4(1 - i)^{-2}$

Q9. Write a short statement to explain when the use of exponential form speeds up calculations for complex numbers and when it is less useful

Extension

Q10. Prove Euler's identity using the Taylors/Maclurins series for sine and cosine

Hint: Use $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \dots$, $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

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